Stage 1 Task Input



Real Video



Reference evidence for evaluation

Stage 2 Agent Reconstruction

Models

Harness

Reference video

coding agent

Revise

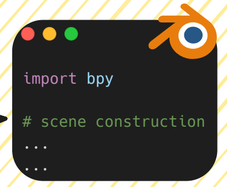


compare



Simulated Video

Code



scene.py

Final submission

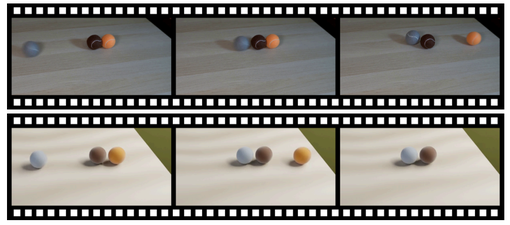
scene.py + manifest.json

Stage 3 Independent Verification

Factual Evaluation

Reference

Reconstruction



LLM as Judge

Object + Event Rubrics

Rule-based Judge

SSIM + LPIPS + Motion

Overall Scores

Visual Score

SSIM | LPIPS | Object

Physical Score

Event | Motion | Validity

Counterfactual Evaluation

Scene intervention

scene.py

Inspect → Copy → Edit

Schematic controls

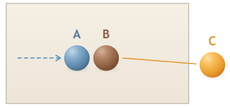
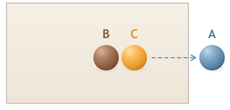
- Ball contact: ON
+ Ball contact: OFF
- Launch field: ON
+ Launch field: OFF

Same initial state
Table support preserved

Re-simulate

Native-state trajectories

t = 3.50 s

Contacts enabled		Target responds
Ball contacts disabled		A moves; C stays
Launch field disabled		No launch

A: incoming ball; B: middle ball; C: target ball

Rule-based Judge

Measured displacement (scene units, t = 3.50 s)

Condition	A	C
Contact on	0.887	0.581
Contact off	1.349	0
Launch off	0	0

✓ Incoming motion preserved
✓ Target response disappears

Contact dependence supported
Scoped mechanism check

Physical validity evidence